

Review 1

The equations of motion of a pendulum of mass m and length L , mounted on a cart of mass M , effectively a Segway, are

$$\ddot{\theta} = \frac{3}{2} \left(\frac{g \sin \theta - \ddot{x} \cos \theta}{L} \right)$$

$$\ddot{x}(M + m) + \frac{mL}{2}(\ddot{\theta} \cos \theta - \dot{\theta}^2 \sin \theta) = 0$$

where x is the horizontal displacement of the cart and θ is the angle of the pendulum to the vertical. Consider the case where $M = 10$ kg, $m = 80$ kg and $L = 2$ m.

- (i) Show that the general solution for the a linear system

$$\mathbf{M}\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$$

about the equilibrium $\mathbf{x}_0$ will be

$$\mathbf{x}(t) = \mathbf{x}_0 + \sum k_i \boldsymbol{\xi}_i e^{\lambda_i t}$$

where the eigenvalues λ_i are the roots of the polynomial resulting from

$$|\mathbf{A} - \mathbf{M}\lambda| = 0$$

and $\boldsymbol{\xi}_i$ is the eigenvector of the i^{th} eigenvalue and k_i is the weighting coefficient for that dynamic mode.

- (ii) Calculate the equilibrium states of the system.
- (iii) For each of the equilibria linearise the system and obtain the general solution for the system's dynamics.
- (iv) Briefly describe the dynamic modes of the system for each of the equilibria.

i. Let $x' = x^* e^{\lambda t}$ where x^* is constant

$$\therefore \dot{x}' = \lambda x'$$

$$\text{At singular point, } \dot{x}_0 = f(x_0, \dot{x}_0) = Ax_0 = 0$$

$$\therefore M\dot{x} = Ax$$

$$\therefore M(\dot{x}_0 + \dot{x}') = A(x_0 + x')$$

$$\therefore M\dot{x}' = \lambda Mx' = Ax'$$

$$\therefore (A - \lambda M)x' = 0$$

$$\therefore |A - \lambda M| = 0$$

$$\therefore x = x_0 + \sum x' = x_0 + \sum k_i \xi_i e^{\lambda_i t}$$

ii. At equilibrium state, $\dot{x}_0 = 0$

$$\dot{x}_0 = \begin{bmatrix} \dot{\theta}_0 \\ \ddot{\theta}_0 \\ \ddot{x}_0 \end{bmatrix} = 0$$

$$\therefore \ddot{\theta} = \frac{3}{2} \left(\frac{g \sin \theta - \ddot{x} \cos \theta}{L} \right) \quad \therefore \sin \theta_0 = 0 \quad \therefore \theta_0 = m\pi, m = 0, 1, 2, \dots$$

$$\therefore \ddot{x}_0 = \frac{g}{L} x_0 \quad \therefore \dot{x}_0 = \text{arbitrary constant} = C$$

$$\therefore x_0 = \begin{bmatrix} \theta_0 \\ \dot{\theta}_0 \\ \dot{x}_0 \end{bmatrix} = \begin{bmatrix} m\pi \\ 0 \\ C \end{bmatrix}$$

$$\text{iii, } M\ddot{x} = f(x, \dot{x}, \delta)$$

$$\therefore M\ddot{x}_0 = f(x_0, \dot{x}_0, \delta) = 0$$

$$\therefore M(\ddot{x}_0 + \ddot{x}') = f(x_0, \dot{x}_0, \delta) + \dot{x}' J_{(x_0, \delta)} + \ddot{x}' J_{(\ddot{x}_0, \delta)}$$

$$\therefore (M - J_{(\ddot{x}_0, \delta)}) \ddot{x}' = \dot{x}' J_{(x_0, \delta)}$$

$$\text{Let } \dot{x}' = x^* e^{\lambda t}$$

$$\therefore |J_{(x_0, \delta)} - I(M - J_{(\ddot{x}_0, \delta)})| = 0$$

$$\therefore M\ddot{x} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{2}{3}L & 0 \\ 0 & 0 & \frac{2(M+m)}{mL} \end{bmatrix} \begin{bmatrix} \ddot{\theta} \\ \ddot{\theta} \\ \ddot{x} \end{bmatrix}$$

$$f(x, \dot{x}, \delta) = \begin{bmatrix} \ddot{\theta} \\ g \sin \theta - \ddot{x} \cos \theta \\ -\ddot{\theta} \cos \theta + \dot{\theta}^2 \sin \theta \end{bmatrix} \text{ while } x = \begin{bmatrix} \theta \\ \dot{\theta} \\ \dot{x} \end{bmatrix}$$

$$\therefore J_{(x_0, \delta)} = \begin{bmatrix} 0 & 1 & 0 \\ g \cos \theta + \ddot{x} \sin \theta & 0 & 0 \\ \ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta & 2\dot{\theta} \sin \theta & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ \pm g & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$J_{(\ddot{x}_0, \delta)} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\cos \theta \\ 2\dot{\theta} \sin \theta & -\cos \theta & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \mp 1 \\ 0 & \mp 1 & 0 \end{bmatrix}$$